



The Inside Story

Here's about how your phone works its magic

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Even a seven-year-old knows how to make and receive calls using a cell phone, but not too many of us know how a phone processes voice and then sends and receives it digitally over the cellular network.

In fact, the question is this: whenever you make or receive a call or text message, what exactly happens in the body of a cell phone? Well, a lot of things: there are several components—and therefore several functions—that work together to deliver what has become very routine. So here's what happens.

1. When you enter numbers via the keypad, a circuit behind it records the key-press sequence as commands in a certain order. These commands are sent to the CPU via firmware located on the ROM.
2. Once the Call button is hit, the CPU sends the sequence of commands to the SIM reader (which reads information from the SIM card). The CPU checks if there is any information about the number on the SIM. The command code is also communicated to other components like the ROM, RAM, and display.
3. Once the process of checking the entered information with the SIM and phone memory is entered, the commands are passed to the DSP (Digital Signal Processor). Along with the voice processing done at the DSP, a highly sophisticated compression of the audio takes place. After that, the processed voice signals are instantly transferred to the transceiver chip in the RF Unit for transmission via the antenna. (A transceiver chip is a combination of transmitter and receiver in a single package for transmission and receiving of signals.)
4. When the cellular base station receives the request for a call, the request is passed

through switches. It is then routed to the destination to check if the destination cell phone is active. Upon this confirmation, the base station routes back the call request to the caller's antenna in the form of modulated signals (radio waves).

5. The cell phone antenna transfers the modulated signals to the transceiver chip (in the RF Unit). RF amplifiers convert the modulated signals to digital instructions, which ride to the CPU.
6. The CPU gives out multiple instructions upon receiving the request. On one side, it gives instructions to the screen to display the relevant information (call being connected, in progress, etc.) On the other side, the OS records the call entry in the "Outgoing Calls" list.
7. Once the call is being connected, the CPU sends the modulated signals to the analogue-to-digital convertors, and the digital voice signals are converted to analogue voice signals that will be sent to the speaker. Ultimately, the dial tone will be audible from the speaker. Once the destination cell phone user presses the Call button, the call request is active and the communication between both the phones begins.
8. When the call is in progress, voice signals in analogue form are sent to the DSP unit according to the instructions of CPU. Those voice signals are processed and converted into digital form by the DSP along with an audio codec. Then, they're instantly transferred to the RF unit. The RF unit amplifiers then process those to transmit via the antenna through the transceiver chip.

The antenna finally transmits the modulated signals on the network in the form of radio waves.

This process sounds lengthy, but it takes just a couple of milliseconds! Let's now take a look at the components of a cell phone.

PCB. Basically, the firmware allows the CPU to talk with the other components of the phone. The phone also carries a customised and compact operating system. Proprietary OSes like Windows Mobile, Palm (only with Palm phones), and Symbian are pre-loaded on smartphones and PDAs.

Keypad

The keypad layout matrix differs from one brand to other. Usually, cell phones have ABCDEF-style keys with numbers on them. When a key is pressed, a command is sent through the rubber- or fibre-made keys to the Keyboard Integrated Circuit for executing the function. Naturally, there's more... but you should have developed a little respect for your mobile phone by now!

Battery

The batteries, too, have a processing mechanism, which keeps track of the charge status and communicates this to the CPU.



The phone's LCD Display

LCD Display

Once the data calculations to be displayed on the screen are processed through the microprocessor, the image gets displayed on the LCD screen connected to the processor. The majority of cell phones in use today use LCD and OLED (Organic Light Emitting Diode) technology for the display screen.

Better displays with lower power consumption is the next step; Hitachi recently displayed its IPS-Pro-Prolleza technology, claiming that display screens will appear 1.4 times brighter than existing displays without increasing power consumption.

Firmware And Operating System

A cell phone's firmware is a hardware-level programming language, mostly a C variant. All major cell phone brands use customised firmware in accordance with the requirements of the hardware. The firmware is hard-coded assembly language that triggers a special user interface and is written on the ROMs of the cell phone's



RF Unit

Whenever radio frequency is transmitted in or out of the antenna, the RF amplifiers in the RF unit handle them. The RF and power section manages the power, recharging, and also FM channels. The frequency ranges allocated to the GSM networks for sending and receiving data over radio waves differ from country to country. The RF unit also communicates with the GSM/GPRS as well as Bluetooth chips for transmitting voice and data respectively.

Microprocessor

A specifically-designed microprocessor that performs all computations in keeping with the features. It is connected with the display and keypad inputs, deals with commands and controls signal with the base station, and also handles computing chores for other functions. Today, besides the main CPU, other microprocessors like Mobile Media Processors are used in cell phones for carrying out multimedia functions (those dealing with digital audio, video, and images). The majority of the cell phones have ARM-based processors along with Digital Signal Processors. Smartphones require faster processors supported by ROM and RAM, as well as Flash memory to boot the phone into the OS.

SIM Reader

The basic purpose of a SIM reader is holding the SIM card so that the information (numbers, SMSes, and other data) on it can be read. When a working SIM (Subscriber Information Module) is inserted in the SIM reader system, a 6-pin mechanism is used for reading the information on the SIM. This is transmitted digitally over the network via RF unit. A similar process takes place when the Call key is hit—the number fed is digitally transferred to the network for call processing.

Digital Signal Processors (behind)

All the voice, SMS, MMS, e-mail, and data/fax features of the cell phone are co-ordinated by the DSP, the most critical unit for signal manipulation. The voice processing by the analogue-to-digital converter is done in tandem with the DSP using either Half Rate (5.6 kbit/s), Full Rate (13 kbit/s), or Enhanced Full Rate (12.2 kbit/s) codecs for identification of the critical portions of the audio, protection, and prioritisation of those parts of signal.

Inbuilt Memory

Various types of phones offer varied functionalities and diverse features, meaning they need different memory solutions. The ROM is permanently on the microprocessor and can run without battery power; the ROM chip initialises the other chips on the circuit board to carry out instructions for running a program. The ROM then copies the boot loader from Flash to RAM and the program is run.

The RAM on the circuit board exists in the form of SRAM, PSRAM or Mobile SDRAM depending upon the mobile design—basic, feature-rich, or smartphone. Their density, system architecture, and read-write operation determines the selection of the memory solution to be used. (RAM is faster than Flash).

Better and faster-to-read Flash memory chips are used in cell phones to hold data like telephone numbers, SMSes, themes, ringtones, images, and videos. The Flash memory serves as the housing for the OS of the phone. Upgrading the firmware means writing an entire new version to the Flash memory. Tasks like transfer phone numbers as well as SMS from internal phone memory to SIM module is possible in majority of the cell phone models.

Speaker and Microphone

Any signals transferred to the speaker and from the microphone have to pass through the Analogue-to-Digital Converter, which communicates with the DSP.

